Building Science

Lecture 11: Water and water management

Dr. Matthew Francis
School of Property, Construction & Project Management
RMIT University, Melbourne, Australia

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Lecture outline

- Categories of Water in Buildings
- Distribution methods of Water in Buildings
- Recycling Opportunities

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Water

Water is necessary for life…

But!

…Water can be a very big problem for buildings!

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Water

What moves in 4 ways:

1. Gravity: (Liquids) flow downhill
2. Kinetic Energy: (Liquids) wind-driven rain
3. Capillary Action: (Liquids) wick through porous materials
4. Air movement: (Vapour) carried by air leaks

A successful building design must resist all four!

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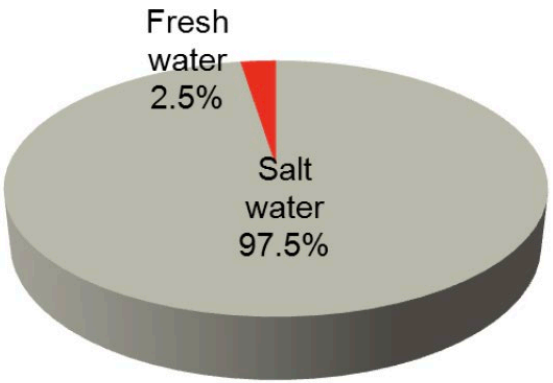
Water

Why is it important?

- ✓ A basic component of human existence
- ✓ One of the most fundamental requirements for the survival of living things



Global water system

- ✓ almost all of the fresh water is in ice or underground
- ✓ the tiny amount (0.01%) is not in ice or underground

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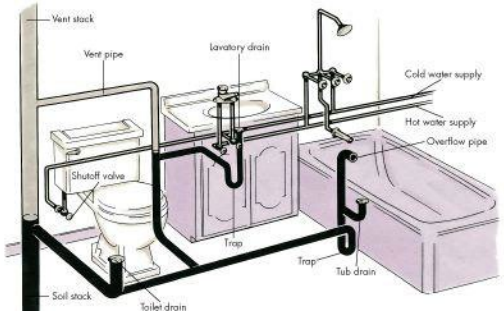
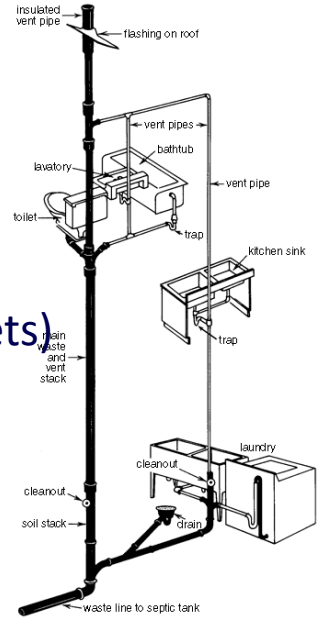
Water



Definitions

- ✓ Potable (or drinking) water: water pure enough to be consumed
- ✓ Grey water: waste water generated from domestic activities such as laundry, dishwashing and bathing.
- ✓ Black water: waste water containing faecal matter and urine (from toilets)
- ✓ Rain water
- ✓ Storm water: water that runs off all urban surfaces such as roofs, pavements, car parks, roads, gardens and vegetated open spaces.



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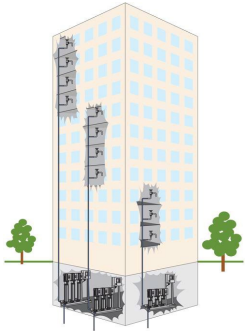
Water

Hydraulic Services Design

What is it? What does this include?

- Sanitary drainage, waste and venting
- Trade waste drainage, plumbing and venting
- Cold and hot water systems
- Rainwater and stormwater drainage
- Gas services



Fire hydrant and hose reel services, with or without pumps
Commercial, residential and domestic fire sprinklers, deluge and wall wetting sprinklers (drenchers)
On-site domestic waste water management.

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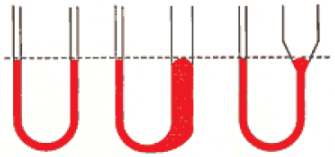
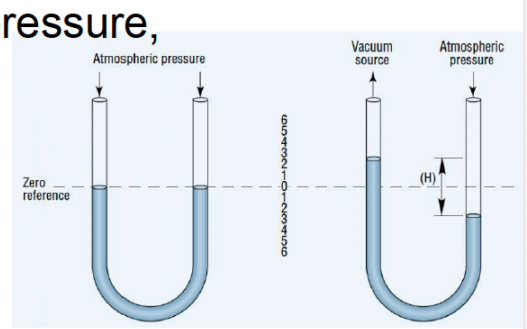
Water

Open Systems

(also known as ‘Open Loop’, or ‘Open Circuit’)



- These systems are ‘open’ to atmospheric pressure, therefore gravity is very important
- E.g. Cooling tower condenser water circuit

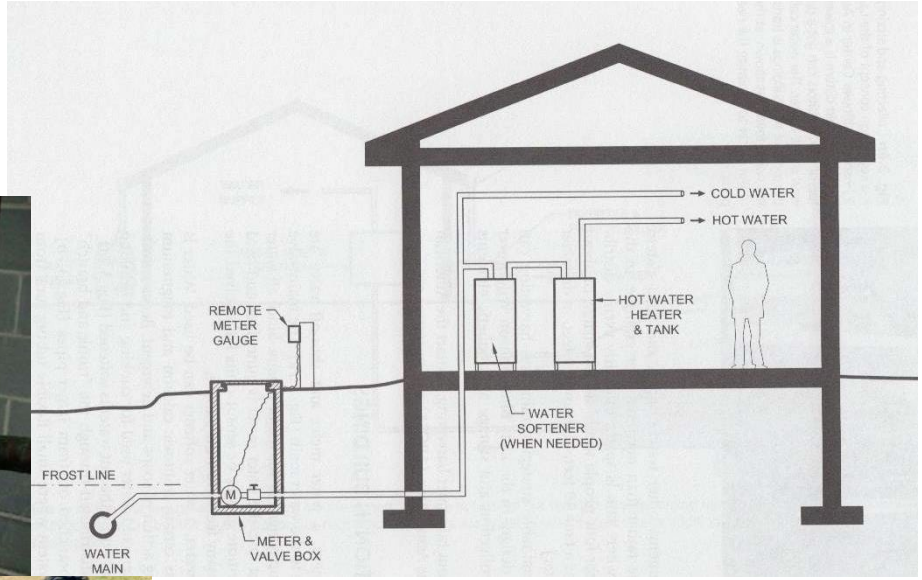


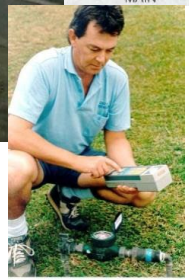
Closed Systems

- Are not ‘open’ to atmospheric pressure
- Direction can be reversed
- Usually under higher pressure
- E.g. Chilled water circuit, Heating hot water circuit.

Water

Water supply





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Water

Water Supply

(1) Water supply from water mains

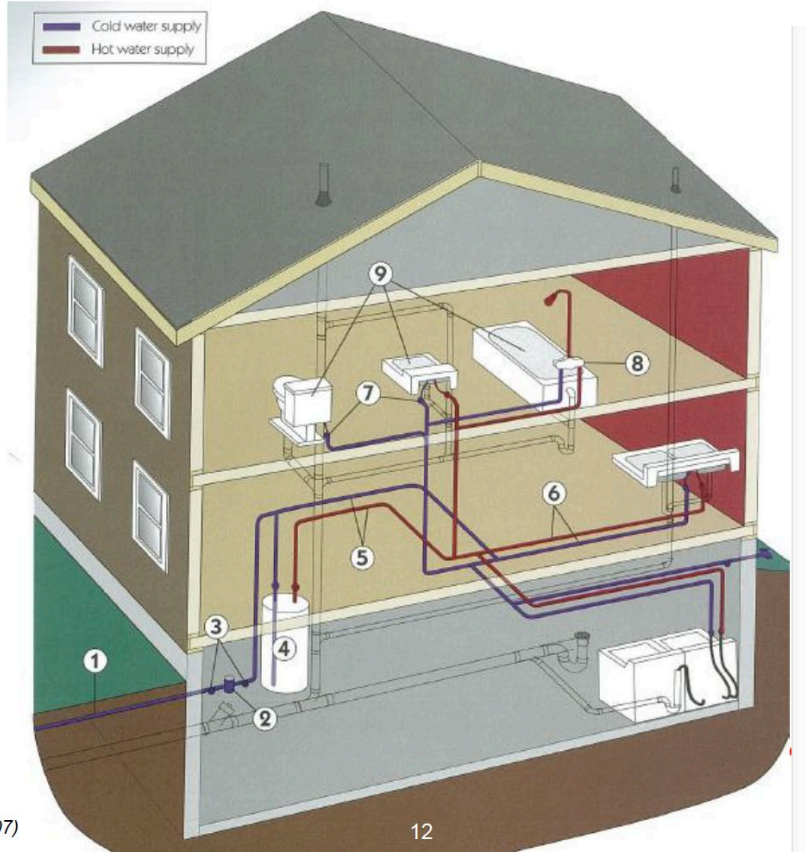
(2) Water meter and valves (3)

(4) Water heater

(5) Supply pipes – cold water and hot water

- *trunk lines*: multiple fixtures
- *branch lines*: 1or 2 fixtures (6)

(9) Fixtures



Source: *C Wing (2007)*

Water

Sanitary Drainage System

(1) Drain

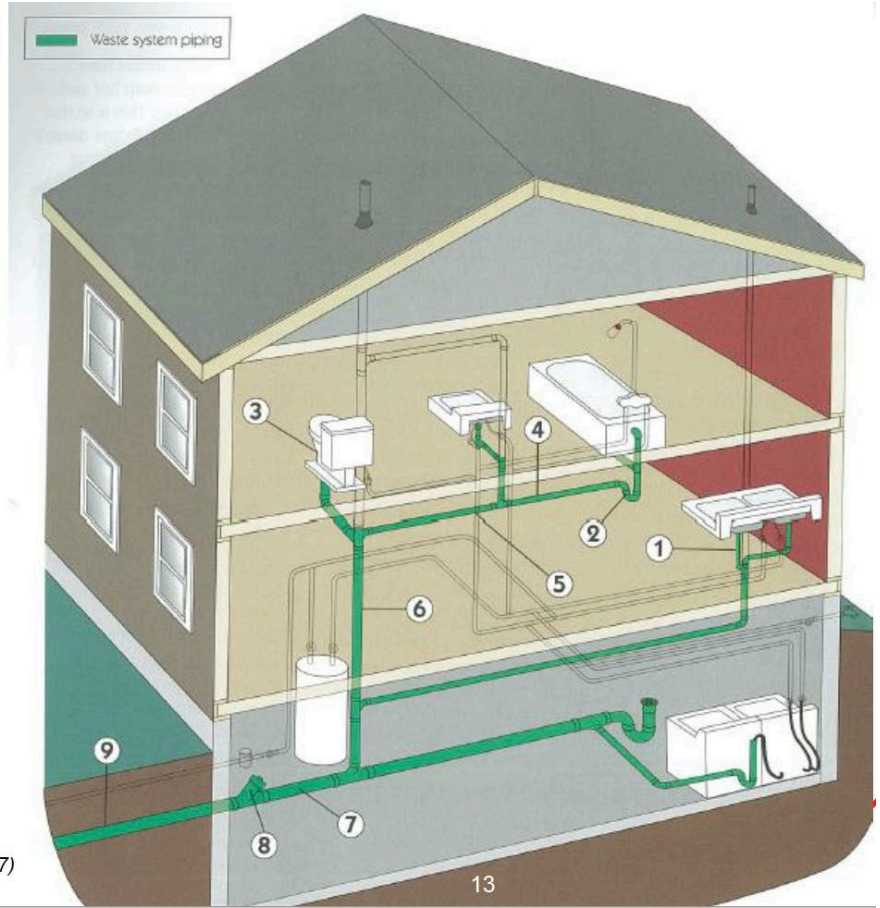
(2) Trap

Horizontal drain pipe
- *trap arm* (4)
- *branches* (5)
- Main *sanitary drain* (7)

(6) Vertical drain pipe
- *stack*:

(8) Cleanouts

(9) Sewer



Source: *C Wing (2007)*

Water

Drainage-waste venting System

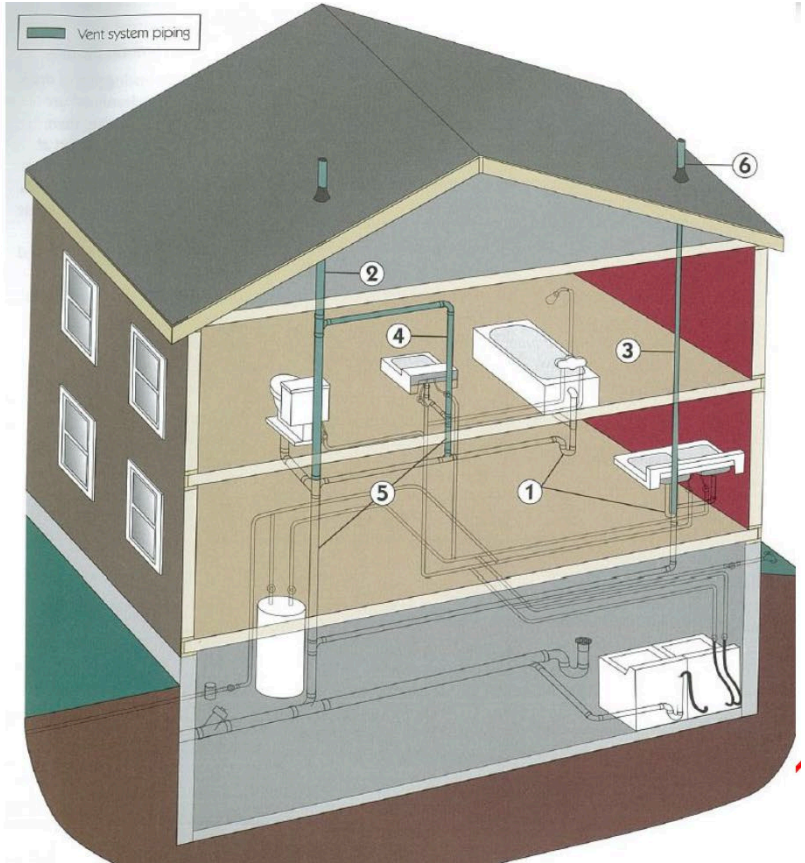
(1) Traps

(2) Primary vent pipe
- *vent stack*:

(4) Revents
- *stack*:

(5) Vertical vent pipe
- *wet vent*:

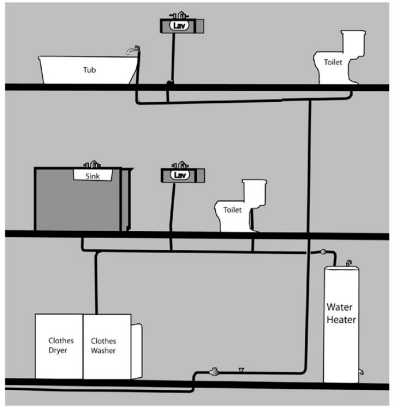
(6) Vent pipes

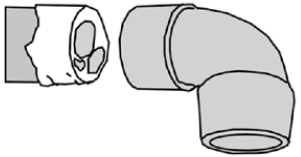


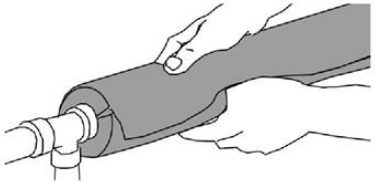
Source: *C Wing (2007)*

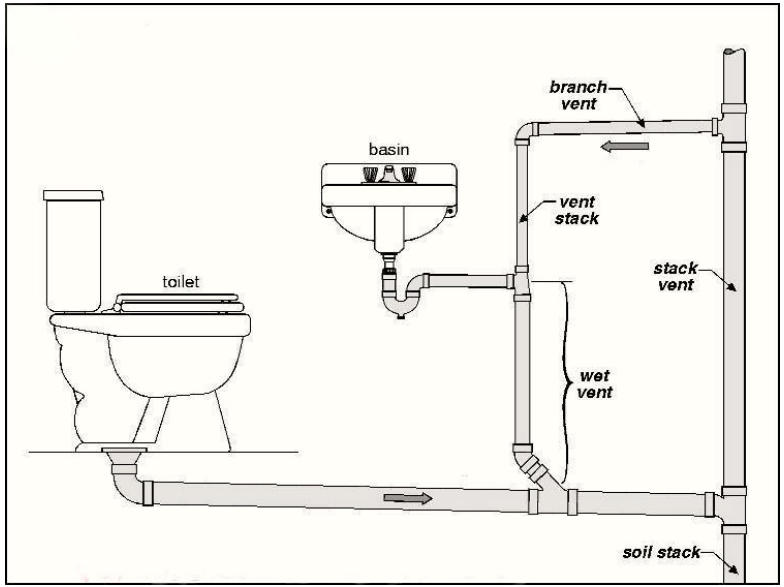
Water

Water distribution









Water

Water distribution

Type	Application	Description
Upfeed distribution	Small low buildings with moderate water use	Water main pressure
Pumped upfeed distribution	Medium-sized buildings with inadequate water pressure	Pumps provide extra pressure
Downfeed distribution	Water raised to rooftop storage tanks; water drops down to plumbing fixtures	Rooftop tanks may need to be heated to prevent freezing

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Water

Water distribution – pipework types

- Piping materials for potable water services:
  - *copper*
  - *stainless steel*
  - *galvanised steel*
  - *Brass*
  - *plastic (PVC, PVDC, PE)*

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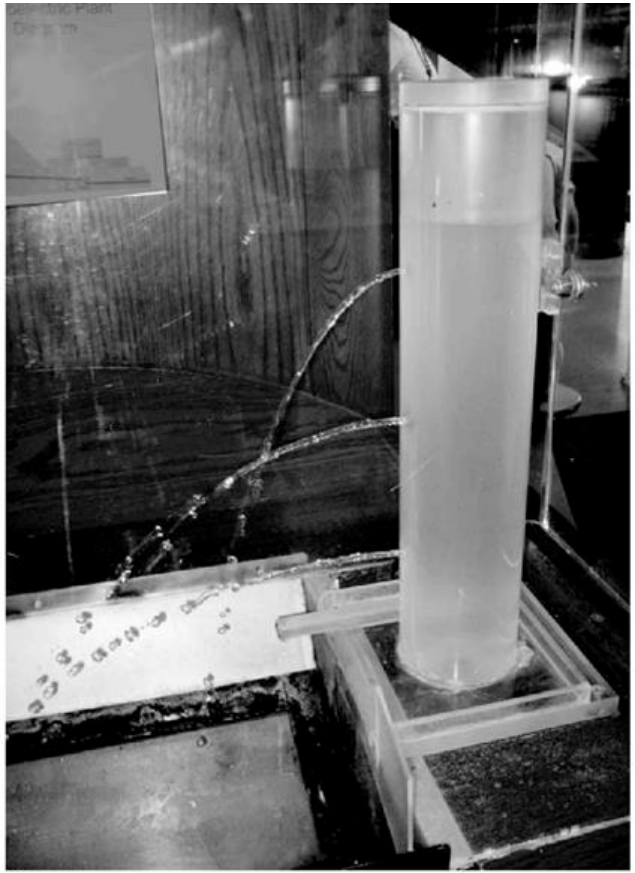
Water

Water pressure:

If a column of water is punctured 3 times, what happens?



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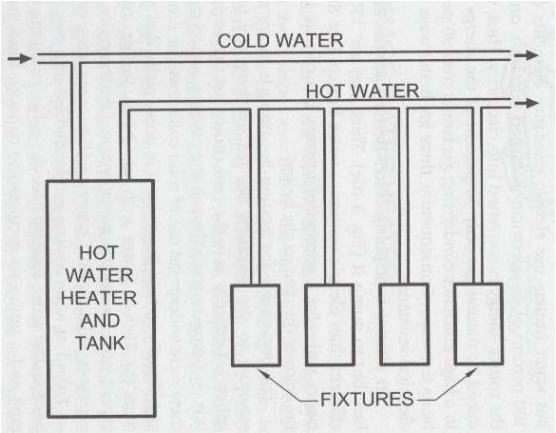
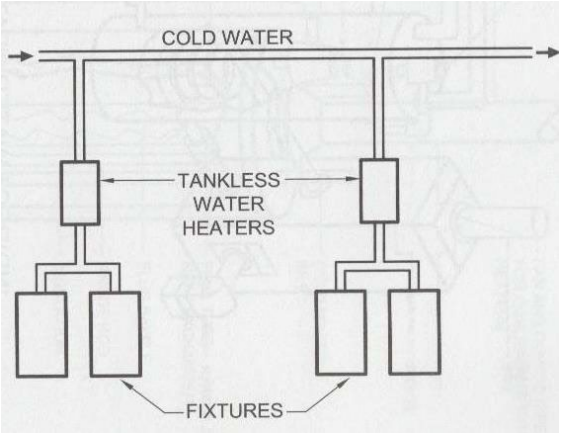
Water

Hot water

Must avoid Legionella disease!





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Hot water temperatures

Activity	Degrees Celsius
Hand washing	40
Shaving	45
Surgical scrubbing	43
Showers, tubs	43
Therapeutic baths	35
Commercial, institutional laundry	82
Residential laundry	60
Commercial spray-type: dish washing	65 (minimum)
: final rinse	82-90

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Waste management

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Waste

Drainage

- ✓ **Drainage**: Pipe work in the ground
- ✓ **Plumbing**: Pipe work in the building above ground
- ✓ **Sewer**: Any conduit belonging to the authority which conveys sewerage or trade waste, away from the site and generally to a sewage treatment works
- ✓ **Soil**: WC, urinals, waste disposal in hospitals
- ✓ **Waste**: Other than soil. Typically basins, showers, sinks. (Grey water)

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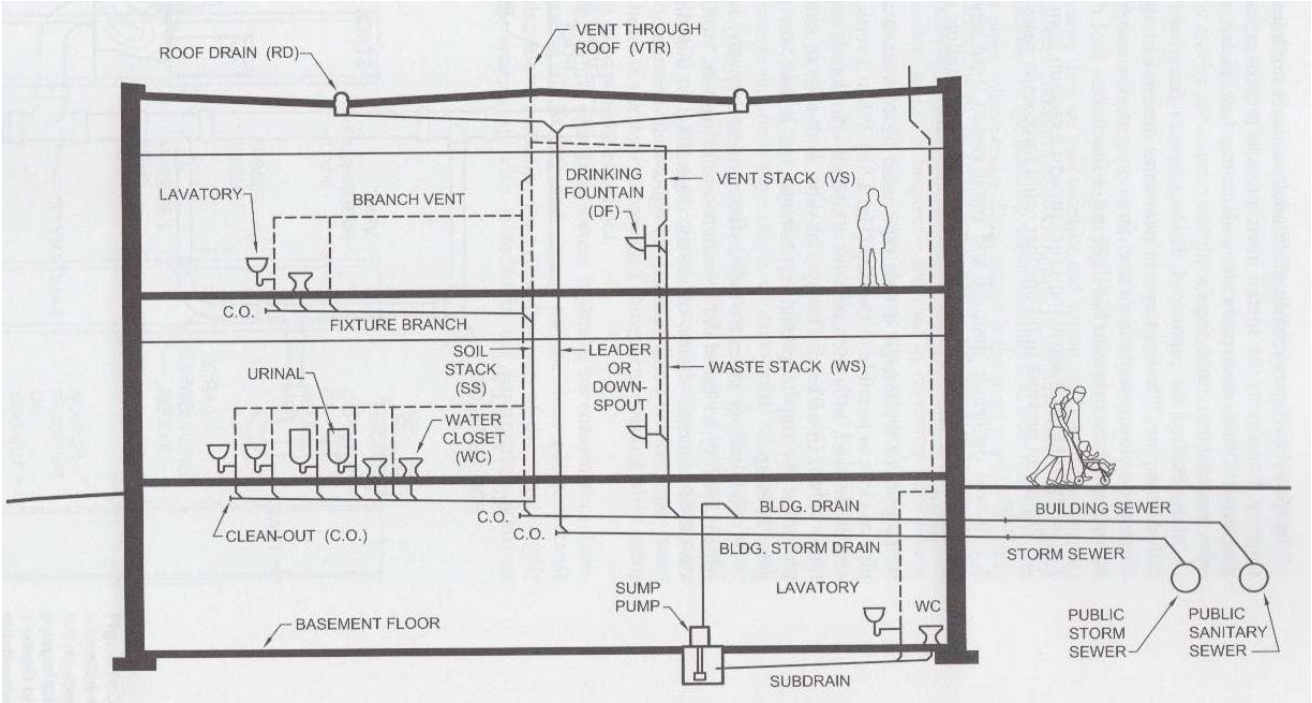
Waste

Drainage piping

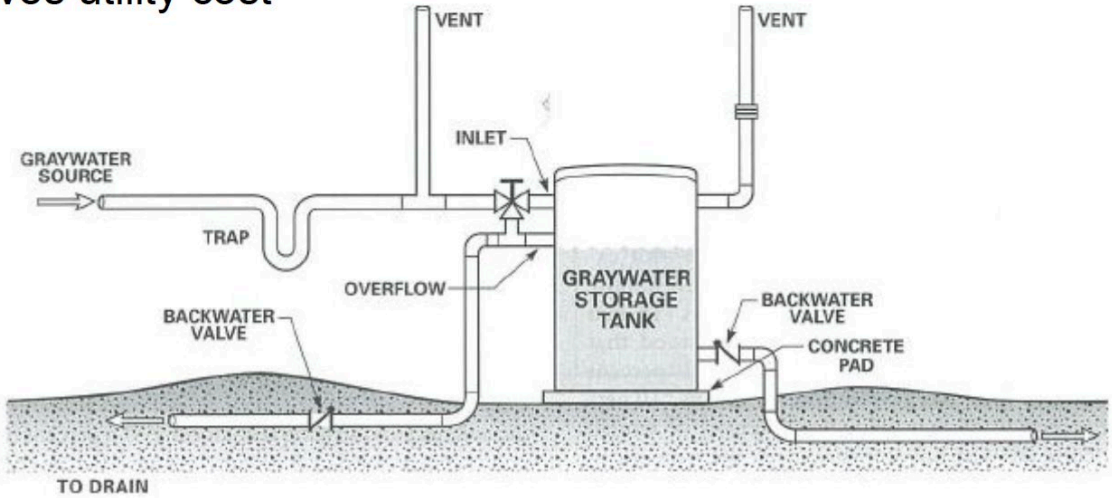


Waste

Grey water systems

Use:

- can reduce overall water consumption
- can reduce sewer load
- saves utility cost



Source: *RR Janis and WKY Tao (2009)*

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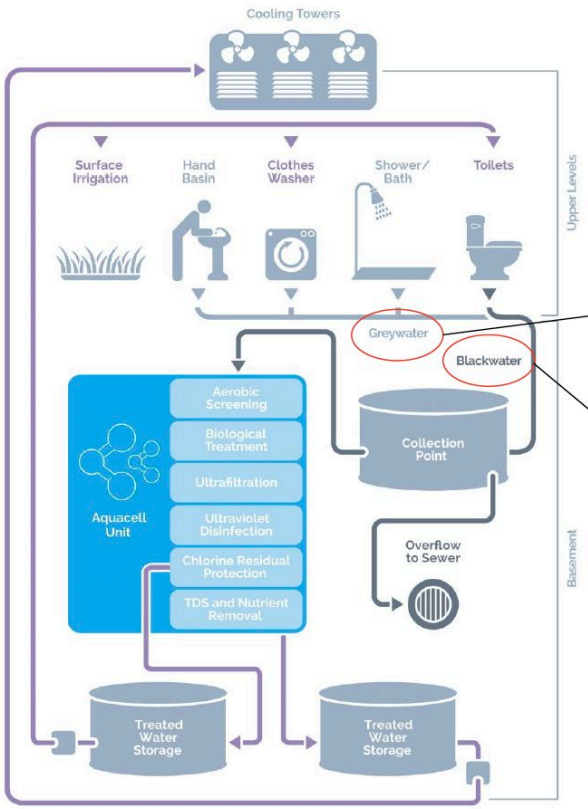
Waste

Black water systems



GREYWATER
- Water that has been used

BLACKWATER
- Water that has been used and contaminated with human waste

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Waste

Black water treatment

BedZED
(Beddington Zero Energy Development)

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Waste

Sustainable building example

Melbourne Water HQ

6-star Greenstar

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Thank you

Next: Final lecture - Energy management in buildings

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